

Local Extrema on Surfaces

In Calc I, a local max of $y = f(x)$ is a "peak" and a local min is a "valley." What does this look like for a **surface** $z = f(x, y)$?

Local Maximum and Minimum

A function $f(x, y)$ has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) near (a, b) .

A **local minimum** at (a, b) means $f(x, y) \geq f(a, b)$ for all (x, y) near (a, b) .

Can a differentiable function f have a local maximum at (a, b) with $f_x(a, b) = 3$?

No! If $f_x(a, b) = 3$, the surface is still *rising* in the x -direction — so (a, b) can't be the highest point nearby. At any local extremum the tangent plane must be **horizontal**: both $f_x = 0$ and $f_y = 0$.

Theorem: Necessary Condition for Local Extrema

If f has a local extremum at (a, b) and the first-order partial derivatives exist there, then

$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0$$

OK, so both partials must be zero. But is that enough? No — a **saddle point** has $f_x = 0$ and $f_y = 0$ but is neither a max nor a min.

At a differentiable extremum, both partial derivatives must be zero — but zero partials alone aren't enough to guarantee an extremum.

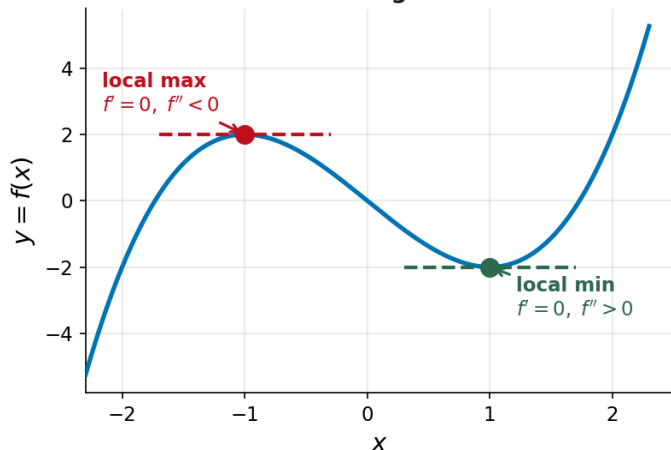
Optimization: From Calc I to Calc III

In Calculus I you learned a systematic optimization procedure. Let's see what changes in Calculus III.

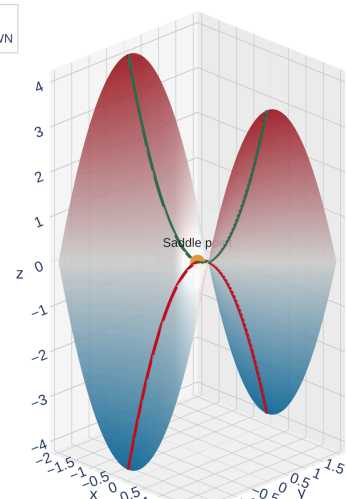
	Calculus I ($y = f(x)$)	Calculus III ($z = f(x, y)$)
Critical points	$f'(x) = 0$ or DNE	$f_x = 0$ and $f_y = 0$ (or DNE)
Classify	2nd Derivative Test	2nd Derivative Test (with D)
New phenomenon	—	Saddle points!
Absolute extrema	Check critical pts + endpoints	Check critical pts + boundary

The big ideas carry over — but the **boundary** is now a curve instead of two endpoints, and **saddle points** are genuinely new.

Calculus I: Horizontal tangent at each extremum



Along x : curves UP
Along y : curves DOWN

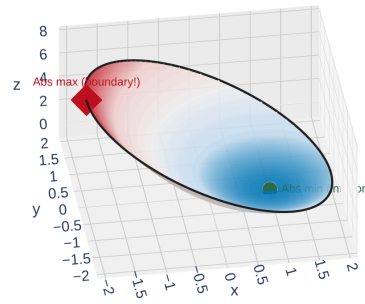
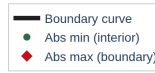
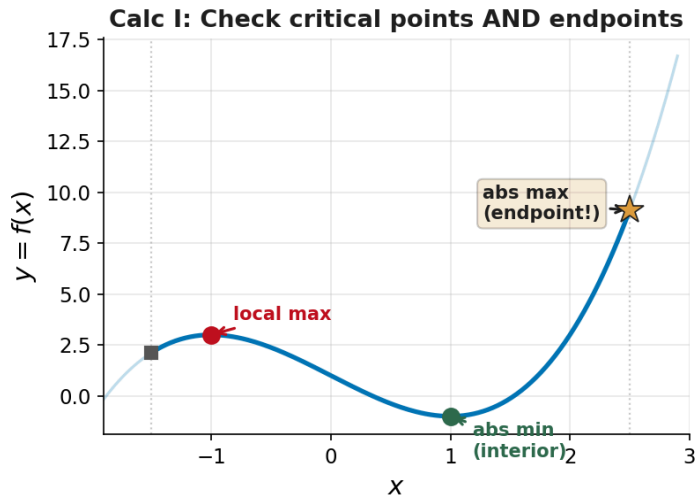


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Familiar: Horizontal tangent ($f' = 0$) at each local extremum. Classify with the sign of f'' .

New in Calc III: At a **saddle point** the tangent plane is horizontal, but the surface curves *up* in one direction and

down in another.



Calc I: The absolute max is at an **endpoint**, not the interior local max. Must check both!

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Calc III: Same idea — but now the “boundary” is a **curve**, not just two points.

Finding Critical Points

Critical Point

A point (a, b) is a **critical point** of f if:

- $f_x(a, b) = 0$ and $f_y(a, b) = 0$, **or**
- one (or both) partial derivatives do not exist there

Every local extremum occurs at a critical point — but not every critical point is an extremum!

Procedure for finding critical points:

1. Compute f_x and f_y
2. Set $f_x = 0$ and $f_y = 0$ simultaneously
3. Solve the system of equations for x and y

Example: Find the Critical Points

Example 1

Find the critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

Compute f_x and f_y , set both equal to zero, and solve the system.

Step 1: Compute partial derivatives.

$$f_x = 6x^2 + y^2 + 10x$$

$$f_y = 2xy + 2y$$

Step 2: Set $f_y = 0$:

$$2xy + 2y = 0$$

$$2y(x + 1) = 0$$

so $y = 0$ or $x = -1$.

Case 1: $y = 0$

$$f_x = 6x^2 + 10x = 2x(3x + 5) = 0$$

$$x = 0 \quad \text{or} \quad x = -\frac{5}{3}$$

Case 2: $x = -1$

$$f_x = 6(-1)^2 + y^2 + 10(-1) = y^2 - 4 = 0$$

$$y = \pm 2$$

Critical points:

$$(0, 0), \quad \left(-\frac{5}{3}, 0\right), \quad (-1, 2), \quad (-1, -2)$$

The Second Derivative Test

Now that we can *find* critical points, how do we *classify* them?

Second Derivative Test

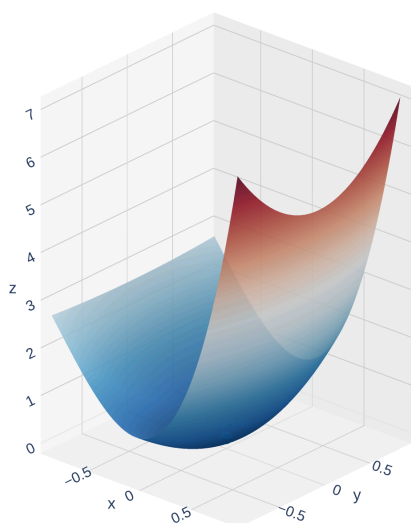
Suppose the second partial derivatives of f are continuous near (a, b) and $f_x(a, b) = f_y(a, b) = 0$. Define:

$$D = D(a, b) = f_{xx}(a, b) \cdot f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

Condition	Conclusion
$D > 0$ and $f_{xx} > 0$	Local minimum
$D > 0$ and $f_{xx} < 0$	Local maximum
$D < 0$	Saddle point
$D = 0$	Inconclusive (test fails)

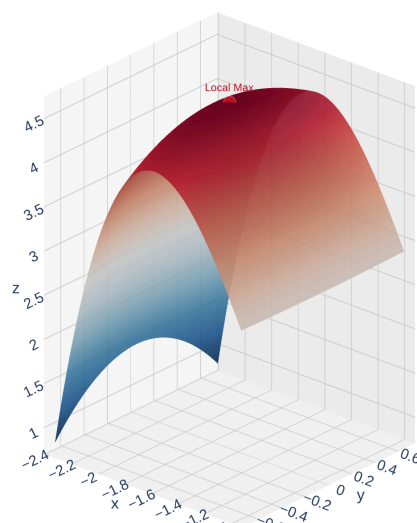
D measures **curvature agreement**: $D > 0$ means the surface curves the same way in all directions. $D < 0$ means f_{xy} dominates — the surface is **twisting**, curving up in one direction and down in another.

Zoomed views of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$ near each type of critical point — **rotate to explore!**



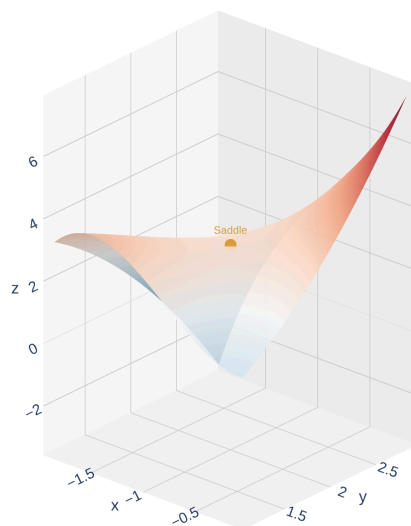
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$D > 0, f_{xx} > 0 \Rightarrow$ **Local min**



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$D > 0, f_{xx} < 0 \Rightarrow$ **Local max**



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$D < 0 \Rightarrow$ **Saddle point**

Quick Check: Use the Second Derivative Test

You've found a critical point and computed:

$$f_{xx} = -6, \quad f_{yy} = -4, \quad f_{xy} = 1$$

Compute D and classify. Local max, local min, or saddle?

Use $D = f_{xx}f_{yy} - (f_{xy})^2$. Check sign of D first, then f_{xx} .

$D = (-6)(-4) - (1)^2 = 24 - 1 = 23 > 0$. Since $D > 0$, it's an extremum.

$f_{xx} = -6 < 0 \Rightarrow$ surface curves **downward** $\Rightarrow$ **local maximum**.

Two-step classification: (1) sign of D — extremum or saddle?

(2) If $D > 0$, sign of f_{xx} — max or min?

Caution: When $D = 0$

Example 2

Show that $D = 0$ at the origin for $f(x, y) = x^4 + y^4$. Is $(0, 0)$ actually an extremum?

Compute all second partials at $(0, 0)$ and find D . Then argue from the formula for f .

$f_x = 4x^3, f_y = 4y^3 \Rightarrow (0, 0)$ is a critical point.

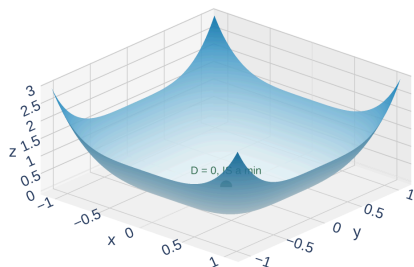
$f_{xx} = 12x^2, f_{yy} = 12y^2, f_{xy} = 0$

At $(0, 0)$: $D = (0)(0) - 0^2 = 0$ — **test is inconclusive**.

But $f(x, y) = x^4 + y^4 \geq 0 = f(0, 0)$ for all (x, y) , so $(0, 0)$ is clearly an **absolute minimum**.

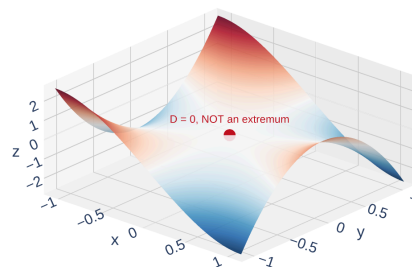
When $D = 0$, you need a **different argument** — the test simply gives no information.

$D = 0$ can go either way! Both surfaces below have $D = 0$ at the origin — **rotate to explore!**



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$f = x^4 + y^4$: So flat that all second derivatives vanish — but still a **minimum**.



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$f = x^3 - 3xy^2$: A "monkey saddle" with three lobes — definitely **not an extremum**.

Example: Classify the Critical Points

Example 3

Classify all critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

(We already found them: $(0, 0)$, $(-\frac{5}{3}, 0)$, $(-1, 2)$, $(-1, -2)$.)

Compute f_{xx} , f_{yy} , f_{xy} , then evaluate D at each critical point.

Second partials: $f_{xx} = 12x + 10$, $f_{yy} = 2x + 2$, $f_{xy} = 2y$

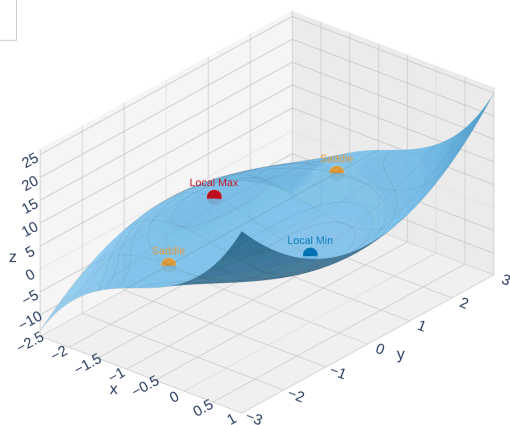
At $(0, 0)$: $D = (10)(2) - 0 = 20 > 0$, $f_{xx} = 10 > 0 \Rightarrow$ **local min**

At $(-\frac{5}{3}, 0)$: $D = (-10)(-\frac{4}{3}) - 0 = \frac{40}{3} > 0$, $f_{xx} = -10 < 0 \Rightarrow$ **local max**

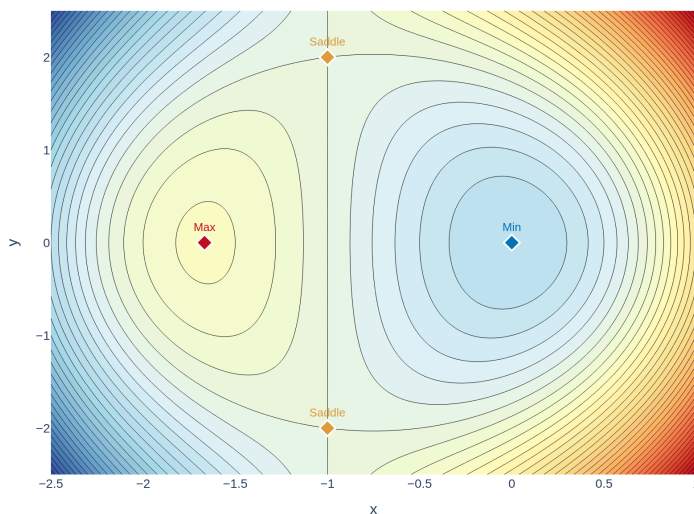
At $(-1, 2)$: $D = (-2)(0) - 16 = -16 < 0 \Rightarrow$ **saddle point**

At $(-1, -2)$: $D = (-2)(0) - 16 = -16 < 0 \Rightarrow$ **saddle point**

- Local Min
- Local Max
- Saddle
- Saddle



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Your Turn: Find and Classify Critical Points

Example 4

Find and classify the critical points of $f(x, y) = x^3 - 3xy + y^3$.

Set $f_x = 0$ and $f_y = 0$. Express one variable in terms of the other and substitute.

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$$

$$f_y = -3x + 3y^2 = 0 \Rightarrow x = y^2$$

Substitute $y = x^2$ into $x = y^2$:

$$x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$$

So $x = 0$ (then $y = 0$) or $x = 1$ (then $y = 1$).

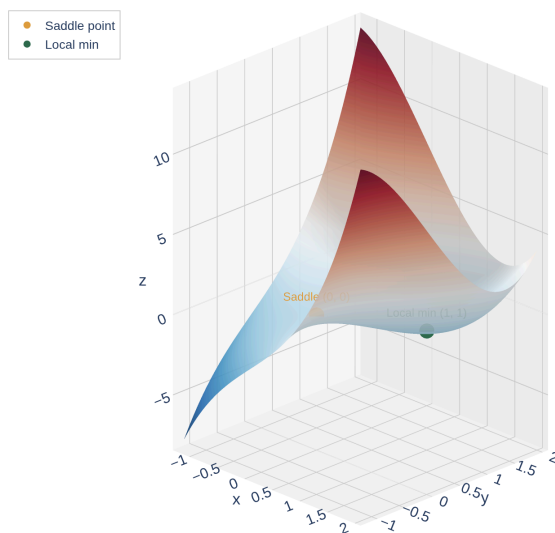
Critical points: $(0, 0)$ and $(1, 1)$.

Classify: $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$

At $(0, 0)$: $D = (0)(0) - (-3)^2 = -9 < 0 \Rightarrow$ **saddle point**

At $(1, 1)$: $D = (6)(6) - (-3)^2 = 27 > 0$, $f_{xx} = 6 > 0 \Rightarrow$ **local minimum**

Two critical points, different classifications. The coupled system required **substitution** — watch for this pattern on homework.



Interactive version

The Extreme Value Theorem

Recall from Calc I: if f is continuous on $[a, b]$, then f attains an absolute max and an absolute min on $[a, b]$.

Extreme Value Theorem (Two Variables)

If f is continuous on a **closed, bounded** set D in $\mathbb{R}^2$, then f attains an absolute maximum and an absolute minimum on D .

Closed set: includes all its boundary points (like $\leq$ vs. $<$)

Bounded set: fits inside some large disk (not stretching to infinity)

Cautionary notes:

- Absolute extrema may occur where f is **not differentiable**
- Absolute extrema may occur **on the boundary** (not just interior critical points)
- There can be infinitely many absolute extrema

Procedure for finding absolute extrema on a closed, bounded set D :

1. Find all **critical points** of f inside D
2. Find the extreme values of f **on the boundary** of D
3. The largest value from steps 1 and 2 is the absolute max; the smallest is the absolute min

Quick Check: Why Check the Boundary?

Why can't we just find interior critical points and pick the largest value?

Where is the absolute max of $f(x) = x^2$ on $[-1, 2]$?

$f'(x) = 2x = 0 \Rightarrow x = 0$ (local **min**, $f(0) = 0$).

The absolute **max** is at the endpoint $x = 2$: $f(2) = 4$.

The critical point gave the minimum — the maximum is on the boundary.

Interior critical points only capture **local** behavior. On a closed, bounded domain, the absolute extrema can occur on the boundary — so we must check both.

Example: Absolute Extrema on a Rectangle

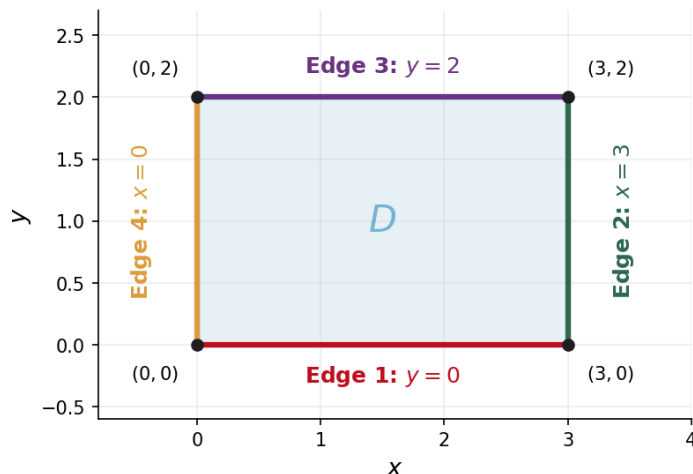
Example 5

Find the absolute maximum and minimum values of

$$f(x, y) = x^4 + y^4 - 4xy + 2$$

on the set $D = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$.

Step 1: Interior critical points. Set $f_x = 0$ and $f_y = 0$:



$$f_x = 4x^3 - 4y = 0 \Rightarrow y = x^3$$

$$f_y = 4y^3 - 4x = 0 \Rightarrow x = y^3$$

Substitute $y = x^3$ into $x = y^3$ to find the interior critical points in D .

Substituting: $x = (x^3)^3 = x^9 \Rightarrow x^9 - x = 0 \Rightarrow x(x^8 - 1) = 0$

So $x = 0$ or $x = 1$ (taking real roots). Gives critical points $(0, 0)$ and $(1, 1)$.

$f(0, 0) = 2$ and $f(1, 1) = 1 + 1 - 4 + 2 = 0$

Checking the Boundary (Continued)

Step 2: Check f on each edge of the rectangle $[0, 3] \times [0, 2]$.

Evaluate f on all four edges. On each edge, one variable is fixed.

Edge 1: $y = 0, 0 \leq x \leq 3$

$$f(x, 0) = x^4 + 2$$

No critical points to find (increasing on $[0, 3]$). Check endpoints:

$$f(0, 0) = 2, \quad f(3, 0) = 81 + 2 = 83$$

Edge 2: $x = 3, 0 \leq y \leq 2$

$$f(3, y) = 81 + y^4 - 12y + 2 = y^4 - 12y + 83$$

Set $\frac{d}{dy}f = 4y^3 - 12 = 0 \Rightarrow y = \sqrt[3]{3} \approx 1.44$. Evaluate:

$$f(3, 0) = 83, \quad f(3, \sqrt[3]{3}) \approx 74.0, \quad f(3, 2) = 75$$

Edge 3: $y = 2, 0 \leq x \leq 3$

$$f(x, 2) = x^4 - 8x + 18$$

Set $\frac{d}{dx}f = 4x^3 - 8 = 0 \Rightarrow x = \sqrt[3]{2} \approx 1.26$. Evaluate:

$$f(0, 2) = 18, \quad f(\sqrt[3]{2}, 2) \approx 11.97, \quad f(3, 2) = 75$$

Your Turn: Check **Edge 4** ($x = 0, 0 \leq y \leq 2$) yourself before looking ahead — it's the simplest one!

Edge 4: $x = 0, 0 \leq y \leq 2$

$$f(0, y) = y^4 + 2$$

No critical points (increasing on $[0, 2]$). Check endpoints:

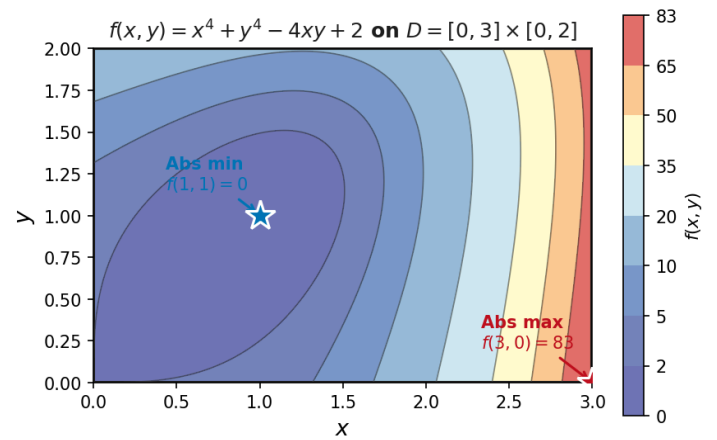
$$f(0, 0) = 2, \quad f(0, 2) = 18$$

Step 3: Compare all values:

Point	f -value
(1, 1)	$0 \leftarrow$ absolute min

$(0, 0)$	2
$(3, 0)$	$83 \leftarrow$ absolute max
$(3, 2)$	75
$(3, \sqrt[3]{3})$	≈ 74.0

Absolute minimum is $f(1, 1) = 0$ and **absolute maximum** is $f(3, 0) = 83$.



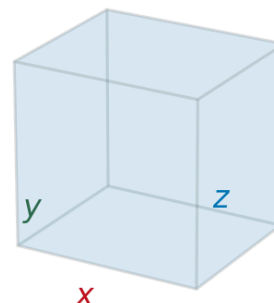
Application: Minimum Surface Area Box

Example 6

Find the dimensions of a rectangular box with volume 1000 cm^3 that has the smallest possible surface area.

Setup: Let the box have dimensions x, y, z (in cm).

- **Constraint:** $xyz = 1000 \Rightarrow z = \frac{1000}{xy}$
- **Minimize:** $S = 2xy + 2xz + 2yz$



Substitute $z = 1000/(xy)$ into S , then find critical points.

Substituting $z = \frac{1000}{xy}$:

$$S(x, y) = 2xy + \frac{2000}{y} + \frac{2000}{x}$$

$$S_x = 2y - \frac{2000}{x^2} = 0 \Rightarrow y = \frac{1000}{x^2}$$

$$S_y = 2x - \frac{2000}{y^2} = 0 \Rightarrow x = \frac{1000}{y^2}$$

$$\text{Substituting the first into the second: } x = \frac{1000x^4}{1000000} = \frac{x^4}{1000}$$

So $x^3 = 1000 \Rightarrow x = 10$. Then $y = 10$ and $z = 10$.

But how do we know this is the global minimum? The domain $x > 0, y > 0$ is open and unbounded — no EVT. Instead, check the boundary behavior:

- As $x \rightarrow 0^+$ or $y \rightarrow 0^+$: $\frac{2000}{x}$ or $\frac{2000}{y} \rightarrow \infty$, so $S \rightarrow \infty$
- As $x \rightarrow \infty$ or $y \rightarrow \infty$: $2xy \rightarrow \infty$, so $S \rightarrow \infty$

S blows up in every extreme direction, so a minimum *must* exist in the interior. There's only one critical point — it wins.

The optimal box is a $10 \times 10 \times 10$ cube ($S = 600 \text{ cm}^2$). On open domains, verify that the function $\rightarrow \infty$ at all edges so the lone critical point must be the global min.

Homework

Section 14.7: 1, 3, 5-21 odd, 25, 33, 37, 41, 43, 45, 47, 49, 51, 55